BITP1113

B032610056

Exercise 1.4

For each of the following problems, identify the problem inputs, processes, and problem outputs. Then write pseudocode and create flowchart for each solution.

1. A program will change a temperature reading from Celsius to Fahrenheit. Prompt the user to enter the temperature in degree Celsius and display the temperature in Fahrenheit. (Fahrenheit = 32 + (9*Celsius/5))

Answer Input : Tempeture In Celsius

Process :

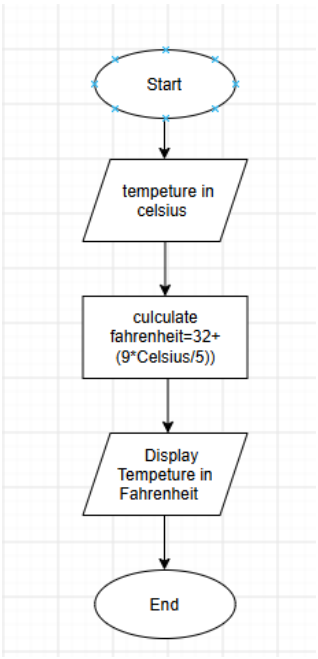
i. Accept Celsius
ii. Culculation from Celsius to Fahrenheit
iii. Display Tempeture in Fahrenheit

Output: Tempeture in Fahrenheit

Pseodocode :

1. Start
2. Prompt user to enter tempeture in celsius
3. Read Celsius
4. Fahrenheit = Fahrenheit = 32 + (9*Celsius/5)
5. Tempeture in Fahrenheit
6. End

Flowchart :



2. Let a runner insert how many steps he has run in one minute, ~~and the running duration in minutes and seconds~~. Find how many miles he has run. Assume each of his step length is 2.5 feet. (1 mile = 5280 feet)

Answer Input :

1. Steps (feet) per minutes
2. Duration in minutes
3. Durations in second

Process :

i. Culculate the duration in miles
   Total minutes =minutes * (seconds/60);
   Total steps = step*total minutes;
   Distance=(total steps*total minutes)5280;

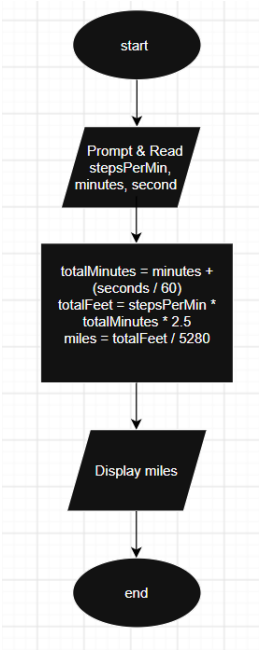
ii. Total minutes = minutes*(second/60);
   Distance in miles = (steps *total minutes*2.5)5280;

Output: Distance In miles

Pseodocode :

1. Start
2. Prompt the user to enter steps perminute
3. Read step per minute
4. Prompt user to enter running duration in minutes
5. Read minutes
6. Prompt user to enter running duration in seconds
7. Read seconds
8. totalMinutes = minutes + (seconds / 60)
9. totalFeet = stepsPerMin * totalMinutes * 2.5 miles = totalFeet / 5280
10. Display "Distance in miles: ", miles
11. End

Flowchart :



3. Accept two numbers and find their sum and difference. If the sum is more than the difference, display the sum otherwise display the difference.

Answer Input : First number , second number

Process :

i. Calculate sum (Sum = num1 + num2)
ii. Calculate difference (Difference = absolute value of (num1 - num2))
iii. Compare sum and difference using an IF-ELSE condition

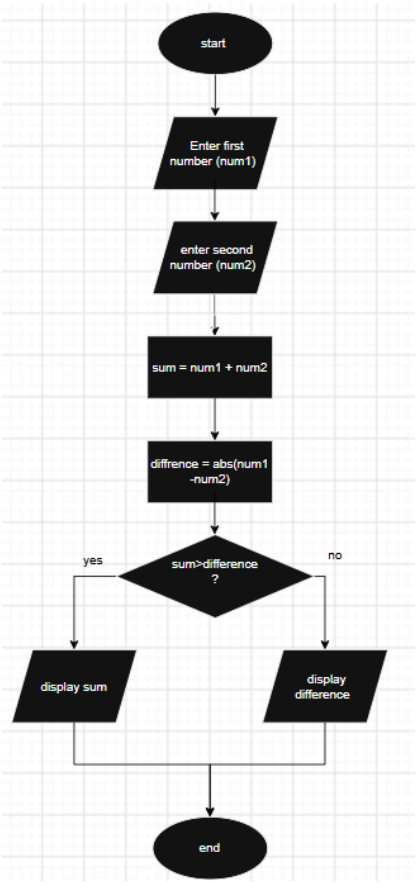
Output: The selected result (Sum or Difference)

Pseodocode :

1. Start
2. Prompt user to enter first number
3. Read num1
4. Prompt user to enter second number
5. Read num2
6. sum = num1 + num2
7. difference = abs(num1 - num2)
8. If sum > difference Then
9. Display "Sum is greater: ", sum

10. Else
11. Display "Difference is greater or equal: ", difference
12. End If
13. End

Flowchart :



4. This is a program to calculate and display the body mass index (BMI). The BMI is to determine whether a person is overweight, underweight or has an optimal weight for his/her height. A person's BMI is calculated using the formula BMI = weight / height2 . A sedentary person's weight is considered to be optimal if his/her BMI is between 18.5 and 25. If the BMI is less than 18.5, the person is considered to be underweight. If the BMI value is greater than 25, the person is considered to be overweight.To determine the BMI, let the organiser enters the total number of people participating in this program. Then let each participant enters his/her weight (in kg) and height (in meters); and as a result, the participant gets a message indicating whether he/she has optimal weight, is underweight, or is overweight. Repeat the process for all participants.

Answer Input :

- Total number of participants (n)
- Weight in kg (for each participant)
- Height in meters (for each participant)

Process :

- Accept the total number of participants
- Repeat for each participant:

  -Accept weight and height

  -Calculate BMI = weight / (height * height)

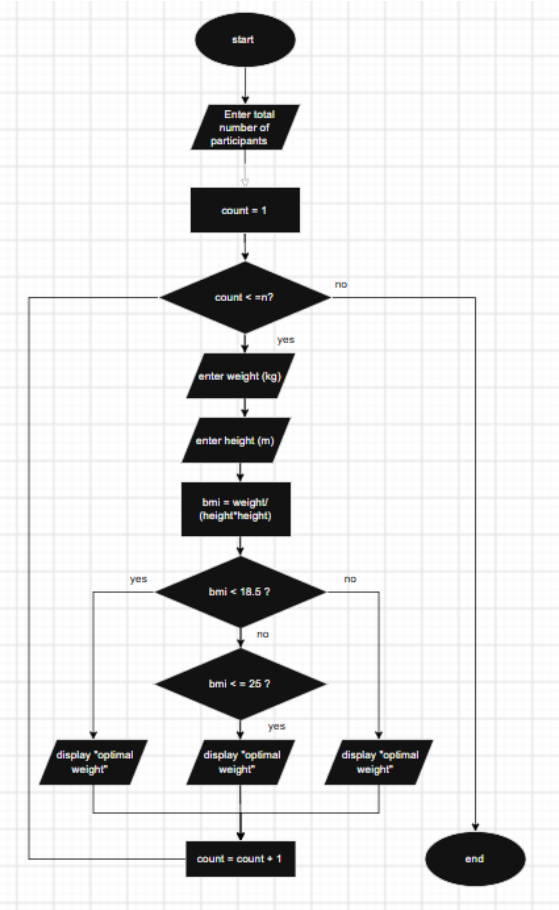
  -Compare BMI: less than 18.5 is underweight, 18.5 to 25 is optimal, more than 25 is overweight

Output: A message for each participant (Underweight / Optimal weight / Overweight)

Pseodocode :

1. Start
2. Prompt user to enter total number of participants
3. Read n
4. count = 1
5. While count <= n Prompt user to enter weight (kg)
6. Read weight Prompt user to enter height (m)
7. Read height
8. bmi = weight / (height * height)
9. If bmi < 18.5 Then
10. Display "Underweight"
11. Else If bmi <= 25 then
12. Display "Optimal weight"
13. Else
14. Display "Overweight"
15. End If
16. count = count + 1
17. End While
18. End

Flowchart :



5. Insert a few integer numbers between -100 and 100 (insert the value 0 to stop). Determine how many negative numbers and how many positive numbers have been inserted, the smallest number and the biggest number, their total and average. Print out what have been determined. (Consider to use the top-down approach in your design).

Answer Input : Input : Number

Process :

If number > 0, increment pos count.
If number < 0, increment neg count.
Track and update smallest and biggest numbers.
Add number to total and increment total count.
Calculate average = total / count

Output: Sum or Diff which ever value is greater

Pseodocode :

1. Start
2. Set Pos Count = 0, Neg_Count = 0, Total = 0, Count = 0
3. Set Smallest = 101, Biggest = -101
4. INPUT Number
5. WHILE Number ?= 0 DO
6. IF Number > 0 THEN
7. ADD 1 To Pos_Count
8. ELSE IF Number < 0 THEN
9. ADD 1 To Neg_Count
10. ENDIF
11. IF Number < Smallest THEN
12. Smallest = Number
13. ENDIF
14. IF Number > Biggest THEN
15. Biggest = Number
16. ENDIF
17. Total = Total + Number
18. ADD 1 To Count
19. INPUT Number
20. ENDWHILE
21. IF Count > 0 THEN
22. COMPUTE Average = Total / Count
23. DISPLAY Pos_Count, Neg_Count, Smallest, Biggest, Total, Average
24. ELSE
25. DISPLAY "No Numbers Entered"
26. ENDIF
27. END
28.

Flowchart :

